

ANALYSTLAB AFRICA

Week 5 Project Summary

HealthConnect Clinic Experience Lab — Data Analytics Track

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1. What I Planned to Accomplish

Building on Week 4's proposed KPIs and business questions, my Week 5 goal was to complete the data quality pass, run a fuller exploratory analysis across all relevant variables, calculate the five proposed KPIs, and produce an initial dashboard with supporting business insights.

2. What I Actually Completed

- Full exploratory analysis across booking lead time, prior no-show history, distance, reminders and reminder channel, appointment type, demographics, and waiting time
- All five proposed KPIs calculated with real figures and interpretations
- A combined visual dashboard (KPI cards + 4 charts) summarising the findings
- Five business insights with direct implications for HealthConnect

3. Key Findings

- Prior no-show history is the strongest single predictor found (43.5% → 68.8% across 0 to 3+ prior no-shows) — stronger than the Week 4 lead-time finding
- Booking lead time remains a very strong second signal (24.7% → 51.5%)
- Reminders have only a modest effect (~4 points), and reminder channel makes little difference
- Distance to clinic matters mainly for the smaller 20+km patient group (57.8% no-show rate)
- Demographics (age, gender) show no meaningful pattern

4. Major Challenges Encountered

The `waiting_time_minutes` field initially appeared to show a striking 66.7% no-show rate for the 60+ minute band — investigating further showed this band contained only 3 records, so the figure was discarded as statistically unreliable rather than reported as a finding. This was a useful reminder to check sample sizes before trusting any cross-tabulation.

5. Important Decisions Made

Cancellations (5.3% of records) were kept as a distinct outcome category rather than merged into No-Show, since they reflect materially different patient behaviour (proactive notice vs. silent absence) and merging them would have diluted the no-show signal.

6. Changes to Week 4 Approach

No changes to the overall approach; the five KPIs proposed in Week 4 were used as planned. The main addition was discovering that prior no-show history is a stronger predictor than any Week 4 hypothesis anticipated.

7. Cross-Track Collaboration Completed

Shared the strongest predictors found (prior no-show history and booking lead time) with the Data Science track's scope, as these are directly relevant to their feature selection and target variable work for the no-show prediction model.

8. Remaining Work

- Deeper dashboard build-out (e.g. in Power BI) beyond the initial static visual report produced this week
- Validating the distance-to-clinic finding with a larger sample if more data becomes available
- Formal business recommendations tied to each insight, to be developed alongside the Data Science track's model results

9. Proposed Focus for Week 6

Translate the strongest findings (prior no-show history, booking lead time) into concrete, testable recommendations for HealthConnect, and build out a more interactive dashboard version incorporating filters by appointment type and reminder channel.